Digital Systems

Lecture 10

Programming with Interrupts

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Following the course of past years by Mike Spivey

Today

? How can we respond to events (that hardware notices), without making the software do nothing while it waits?

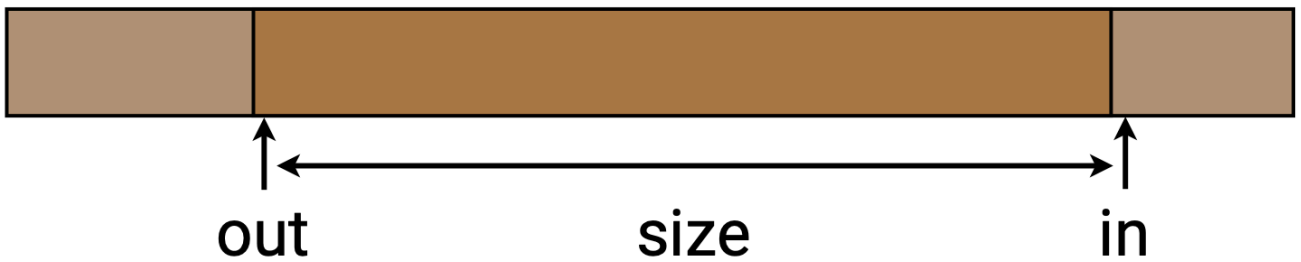
We will consider various solutions, keeping in mind *design constraints*.
We want:

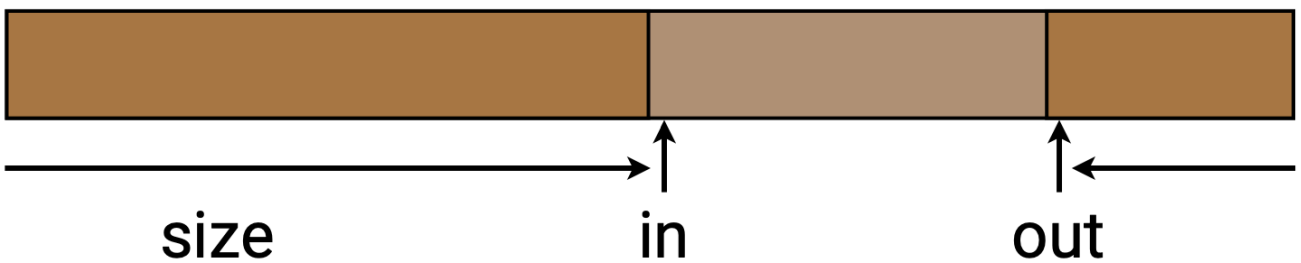
- to avoid making the CPU wait until some event occurs in hardware,
- to be able to deal with many different types of events (not just UART, but e.g. also noticing when a button is pressed),
- to have a minimal waiting time to respond to an event,
- software that is modular, in the sense that it doesn't need to be adapted when other parts of the program change which events they deal with.

One Possible Solution

- `serial_putc()` no longer waits and sends character.
- Instead, it places a character in a buffer.
- Throughout the program, check if UART is ready, and send character if so.

Circular Buffer





- Add characters at `bufin`.
- Send characters from `bufout`.

One Possible Solution

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) { pause(); }   // Buffer is full...
  txbuf[bufin] = ch; bufcnt++;
  bufin = (bufin+1) % NBUF;
}
```

```
int prime(int n) {
    int k = 2;

    while (k * k <= n) {
        if (n % k == 0) return 0;
        poll_uart();
        k++;
    }

    return 1;
}
```

```
void poll_uart(void) {
    if (UART_TXDRDY) {
        // send another char
    }
}
```

What do we think of this solution?

- ✅ to avoid making the CPU wait until some event occurs in hardware,
- ❓ to be able to deal with many different types of events (not just UART, but e.g. also noticing when a button is pressed),
- ❓ to have a minimal waiting time to respond to an event,
- ❌ software that is modular, in the sense that it doesn't need to be adapted when other parts of the program change which events they deal with.

Better Solution: Interrupts

- When `UART_TXDRDY` becomes true, execute `uart_handler()`.
- Variables that are shared between interrupt and main code, must be marked `volatile` (can change any time, outside direct code execution).

Better Solution: Interrupts

```
void uart_handler(void) {
  if (UART_TXDRDY) {
    UART_TXDRDY = 0;
    if (bufcnt == 0) {
      txidle = 1;
    } else {
      UART_TXD = txbuf[bufout];
      bufcnt--;
      bufout = (bufout+1) % NBUF;
    }
  }
}
```

Better Solution: Interrupts

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) pause();//volatile ensures fresh read
  intr_disable();
  if (txidle) {
    UART_TXD = ch;
    txidle = 0;
  } else {
    txbuf[bufin] = ch; bufcnt++;
    bufin = (bufin+1) % NBUF;
  }
  intr_enable();
}
```

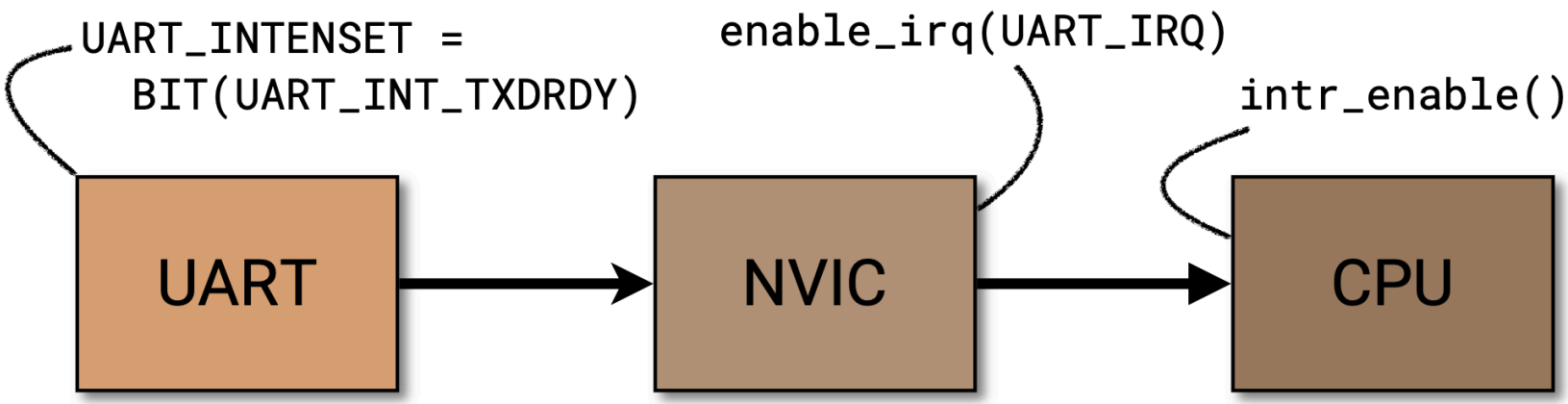
Why Disable Interrupts?

- What happens if interrupt occurs before `bufcnt++;`?
- What happens if interrupt occurs

```
ldr r0, =bufcnt
ldr r1, [r0]
                        @ <== here?
add r1, r1, #1
str r1, [r0]
```

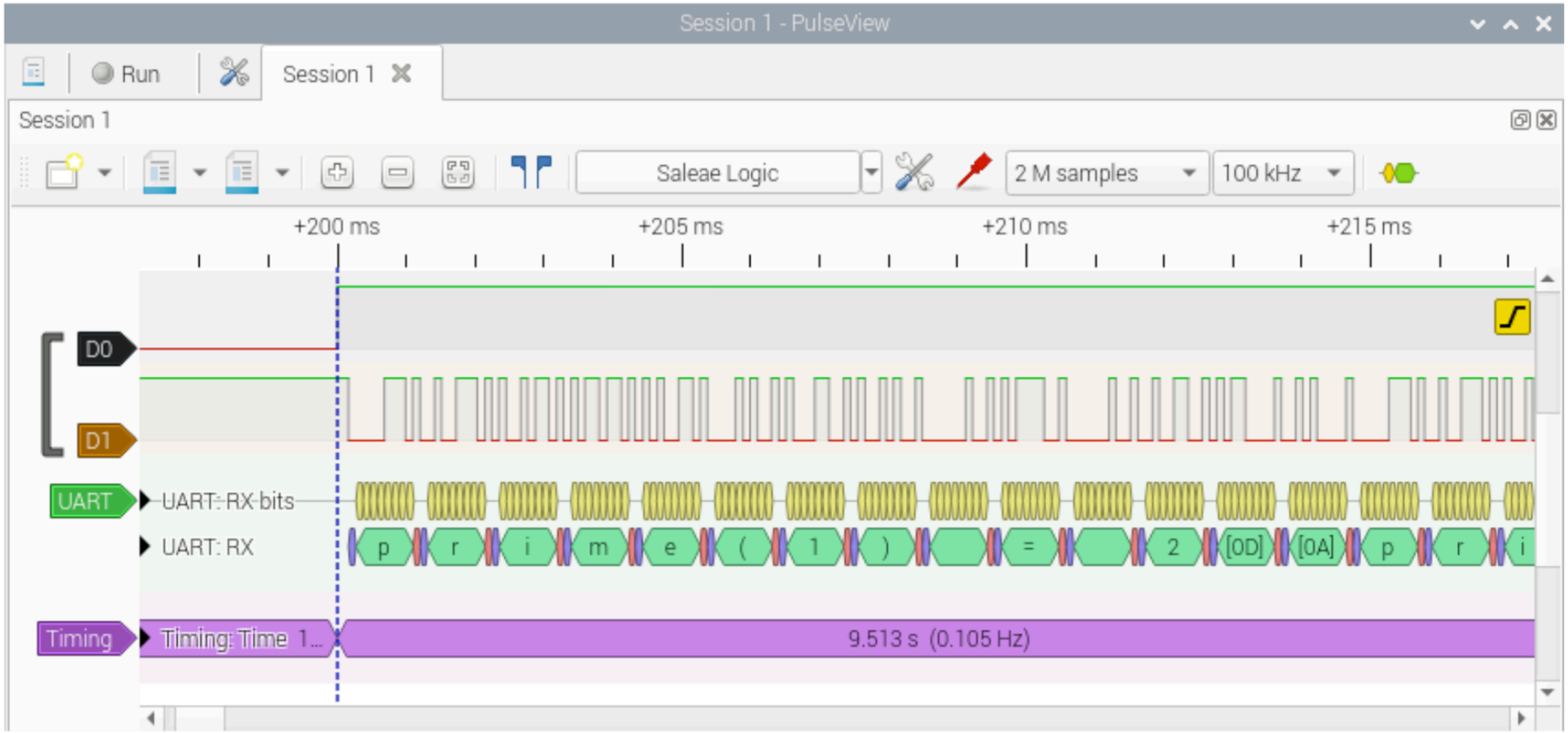
Setting Up Interrupts

```
void serial_init(void) {
  ...
  UART_INTENSET = BIT(UART_INT_TXDRDY);
  enable_irq(UART_IRQ);
  txidle = 1;
}
```



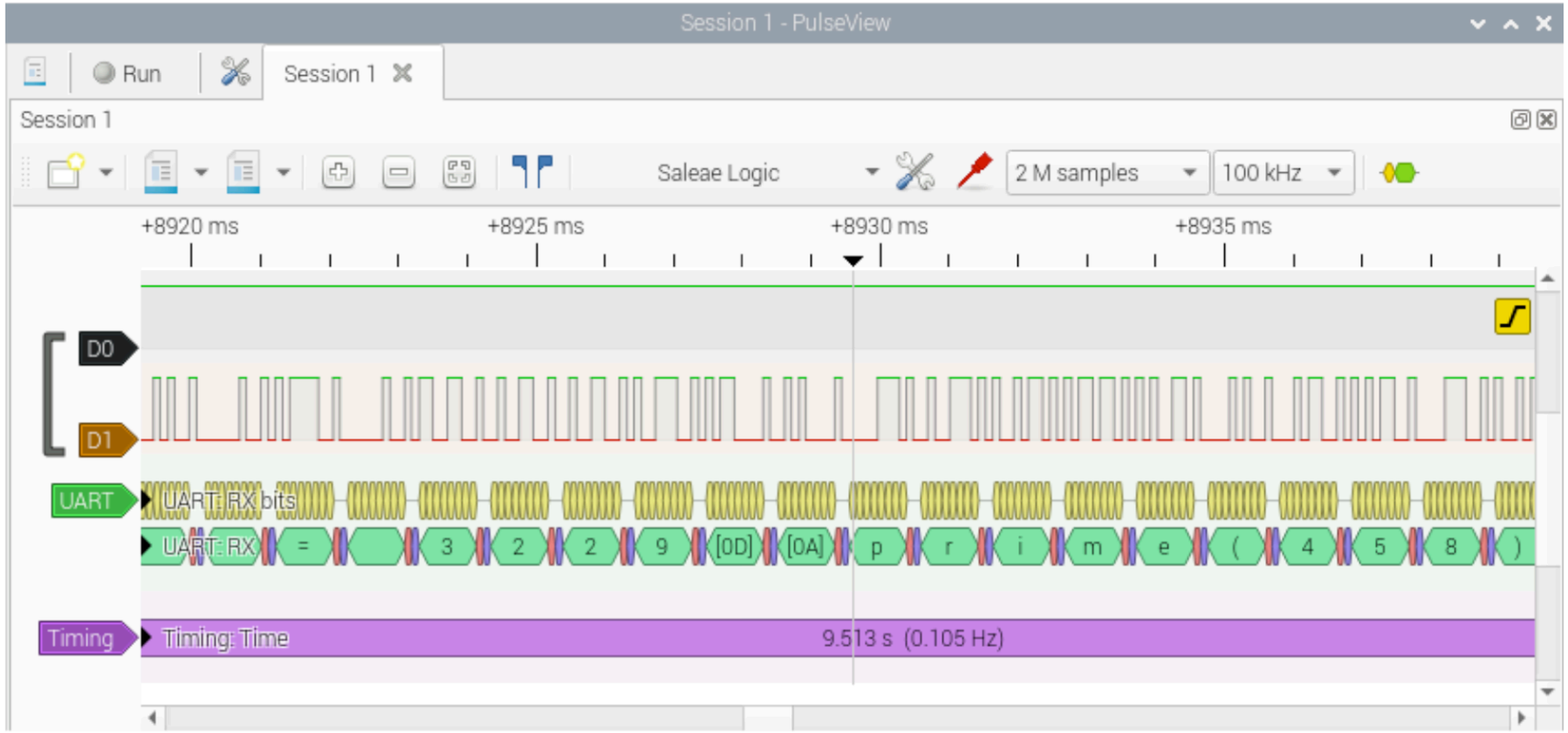
Timings: Early on in Primes Program

No (noticeable) delays between sending characters.



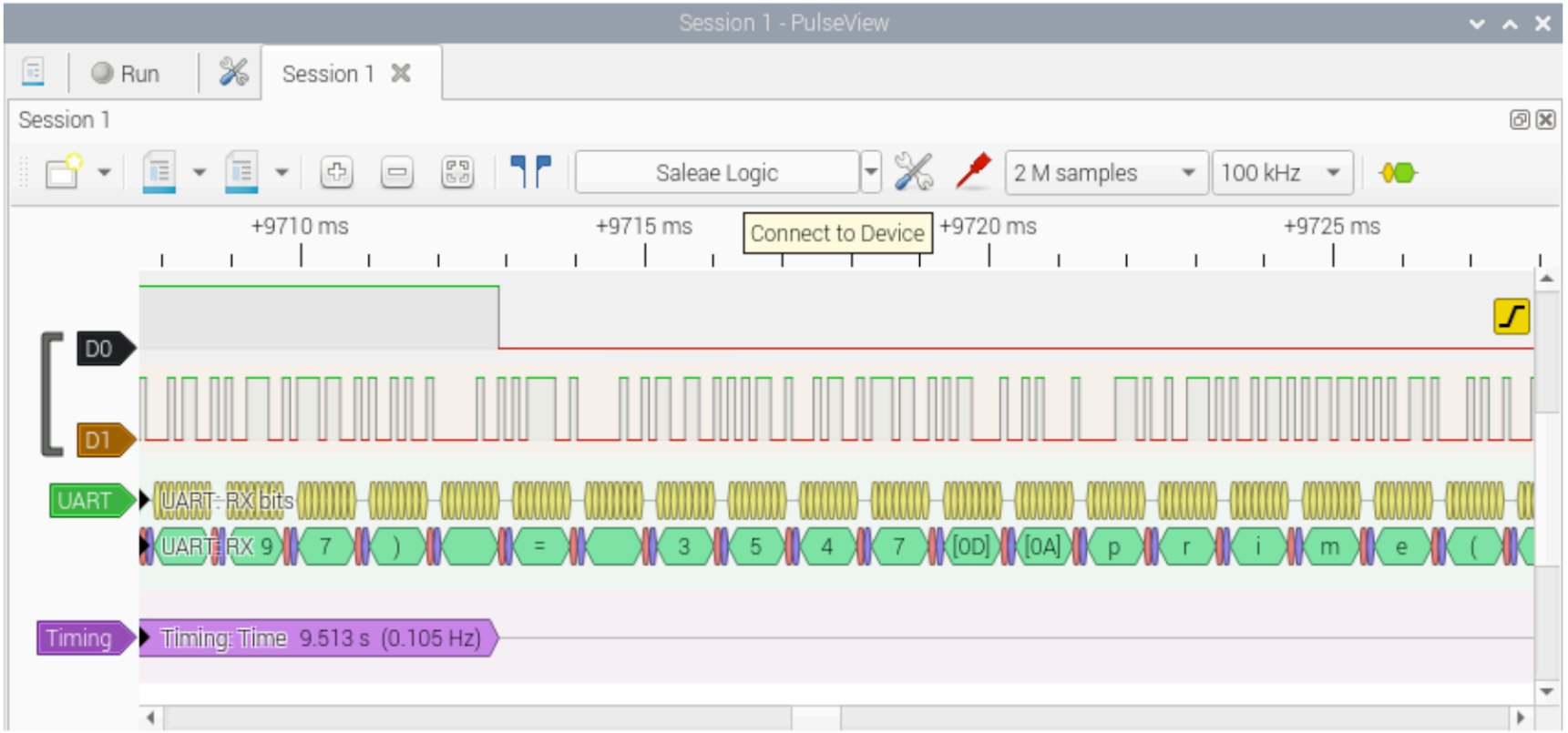
Timings: Later on in Primes Program

No (noticeable) delays between sending characters.



Timings: End of Primes Program

Sending characters continues after primes calculation is done!



⚠ **Crucial detail omitted!**

Summary

- How to implement a queue with a circular buffer.
- Why we need interrupts.
- What hardware provides for interrupts.
- How to use interrupts in software.